

# Visualising Natural Disaster Image Embeddings

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## **1. Introduction**

Someone who becomes a victim of natural disasters or witnesses it will use social media to post information about it and the aftermath. Just after a crisis event, the internet specifically social media becomes a main hub for all types of data that would lead to real time situational data and decision making. Having accurate data and information is important for good humanitarian response. Social media platforms like X and Instagram will have loads of important data about it such as videos, images, audios and text messages from people present nearby. Situational Awareness is able to show disaster management personnel and other humanitarian workers about the aftermath of the natural disaster but in reality, this can be very difficult. There is just too much information and data for any human to look at. Datasets online can be unstructured and is probably referred to as “Information noise”. Imran et al. (2014) says that during important events, there are lots of irrelevant or unwanted data out on the internet that can prevent suitable response actions when dealing with major life threatening events which makes raw streams useless. Not just that there is a research paper from Castillo (2016). The authors argue that without automated filtration, the high signal to noise ratio on the internet affects decision making instead of helping it.

This is a problem nowadays and to address this challenge, a dataset for training an AI model was created. This dataset is called “CrisisMDD” (Alam et al., 2018). This dataset is very important because it kind of made a breakthrough in how we can get computers to process multiple information much quickly and this dataset also has multimodal data of images and tweets that are annotated specifically for humanitarian tasks such as locating infrastructure damage and people that need rescuing. It helped use to check whether multimodal data would perform better than just image data alone.

This might sound cool but there is another problem I have to discuss first. Majority of the previous works done on this same database including the benchmarks created by Ofli et al. (2020) used Supervised Learning where the model is trained to hunt down specific predefined targets such as “Fire” or “Flood”. Even though this could all work very nicely in a laboratory environment, the AI model is not very adaptable and useful within real world environments and conditions where probably like a new kind of natural disaster happens which is not recognised by the AI system. I would also say that labelling thousands of images by humans within a real-time environment isn’t possible. The issue in this project is that we must break from these very structured, supervised processes of classification. What is needed the most is the technology equipped for unsupervised Exploration which is an infrastructure capable of placing similar images into similar groups without needing to first define what even those images are. Using a multimodal learning specifically CLIP (Radford et al., 2021) it could probably be possible to construct a technology that is capable of a semantic understanding of disaster images, where a search can be used for what is needed rather than a pre-structured checklist.

## **2. Visual Feature Extraction**

In the last 10 years, the paradigm in Computer vision has been very simple. If you wanted an AI model that could detect a flooded street then you would need to implement a Convolutional Neural Network structure for example the one found in the ResNet model. These models are really good at what they can do but they function within a rather hard box. These models learn by examining tons of images in the thousands that are labelled “Flood” and detecting the particular pattern of pixels that correspond with “Water”.

I have also considered using two alternatives for feature extraction in image classification tasks called ResNet-50 and Convolutional Autoencoders. Even though there is no doubt that ResNet-50 is still currently considered industry-standard within image classification by professionals, it is also entirely ‘language blind’. What I mean by that is that it can say if there’s “water” in an image, but it would not know whether it is “flood water” or “lake water” which is a problem for me and not just that it would need to be heavily re-trained again.

The bottleneck is well known. Voulodimous et al. (2018) says that deep learning models that are trained on specific datasets do fail to generalise to scenarios that the dataset doesn’t include because of domain shift. If we’re talking about disasters, Nguyen et al. (2017) has shown that standard classifiers do actually struggle with the visuals of the damage and sometimes misclassifying things like background rubble as infrastructure damage which is not true and this happens because of semantic context.

I have also thought about using Autoencoders to lower my images to latent vectors. Even though this would have made clustering to be functioned in an unsupervised way, autoencoders are just learning pixel-level characteristics for example like edges and texture and do not learn at a deep and abstract level. I considered not using such uni-modal models in favour of CLIP because it is the only model that has managed to fix the problem of semantic gap between visual data and natural language inputs.

It is at this point that the method and approach from Radford et al. (2021) is a breakthrough for how we can complete this task. Their approach is called Contrastive Language-Image Pre-training (CLIP) skips the traditional approach of training based on fixed labels and it is also described as Multimodal, which means that it trains on images and text at the same time. CLIP can pair visual scenarios with their semantic meaning because it was trained on 400 million pairs of images and text from the internet. They don’t only just recognise a shape but also it can link the image to the actual text description of any given image input. The huge benefit that this method will provide and help in my project is known as Zero-shot learning. This is because the relationship CLIP understands the relationship between language and the visuals which is then able to identify photographs that falls outside of its technically trained parameters that wasn’t used in training or used in the dataset and can still relate to anything given to it.

### **3. Dimensionality Reduction and Visualisation**

So now that the CLIP model has analysed the images we are not done yet because we still have another problem to fix first before concluding. The CLIP model gives me “Embeddings” which are lists of digits that describe the semantic content of each images separately. These embeddings are expressed in a 512 dimension vector space. The truth is that we can only visualise a maximum of 2 or 3 dimensions. This means that we require a mathematical solution to “Squash” the vectors onto the 2D screen. This is called Dimensionality Reduction and I will be using this method in my project.

When I was thinking about what algorithms to use for dimensionality reduction, the first method that came up to my mind was the Principal Component Analysis or PCA for short. This is the oldest statistical technique for projects like this, this is because it has an advantage of being very quick in speed computationally. But however, I cannot use this as it is a linear method only.

The data from the images is never in a 2D plane but in a curved non-linear manifold instead. If I were to try and visualise it in 2D with PCA, I would have to like “smash” the data together. The points that were actually far apart would end up on top of each other, removing the semantic meaning. Also, PCA would also have a reduced the complex because the data in the CrisisMDD dataset has complex, non-linear relationships between other disasters and subtle clusters into something that wouldn’t have meant anything in the visualisation. This is why I chose the UMAP algorithm in my project because it works well with the manifold in my dataset.

I then started studying non-linear graphs of neighbours, focusing on the application of the t-SNE method. As it will become clear when explaining the non-linearity.

The default method for this process was t-Distributed Stochastic Neighbour Embedding (t-SNE) which is still used in many other use cases depending on the task. Actually if you take the average Data Science guide, t-SNE is used a lot and still is common today. This is particularly good at detecting local clustering and for example, would place photos side-by-side. The fundamental problem about this algorithm is that it is massively expensive compute-wise. This is because the bigger the dataset, the longer it will take to perform calculations. Even though the CrisisMDD dataset is around 18,000 photos, using t-SNE algorithm would definitely result in the program unresponsive and useless which destroys the interactive part of my program.

But however, t-SNE suffers form a structural problem. It is purely based on “local structure” without worrying about “Global structure”. This implies to me that even though the clusters appear pretty, it does not make any sense to measure distance between them. It can happen in a t-SNE graph that a set of “Floods” can appear alongside “Earthquakes” simply by probability without any relevance between them.

This is the reason why I have decided to use Uniform Manifold Approximation and Projection which is as described by McInnes et al. in 2018. The reasons I have chosen UMAP are that it fixes both of the issues I mentioned and described above. It is much faster than t-SNE which means that I can perform transformations on the entire set of Images that I have much quickly probably in a matter of hours, rather than days or even months for t-SNE. However, UMAP is designed with the goal of preserving the global structure of the data which is really good. This means that the actual visualisation is meaningful. If the “Flood” points are plotted away from the “Fire” points then this is an actual semantic success for me. This is very important for my project because while I want to have a nice graph, I want to have something that I can use to actually analyse the relationship between the points in a better way for a responder to analyse the disaster scenes and data.

Having visual tech is very important and it is supported by Andrienko et al, (2007). He says that geovisual analytics is important for helping humans find complex patterns. Liu et al. (2008) says interactive visual interfaces aren't aesthetic but requirements for maintaining situational awareness under pressure in crisis management.

## **4. Conclusion**

I would say that there is definitely a bottleneck in modern technology use for disaster response today. We have the availability of massive amounts of data for example CrisisMDD, but our capability for real-time filtering is constrained by the inflexibility of conventional models for disaster response technology based on normal Supervised Learning algorithms. Like for example, even very powerful CNNs are not even flexible enough to capture dynamic nature of a disaster scene for disaster response technology.

I have found that there are two technologies that could work and be made possible in the field. The first one is CLIP which has no need of labels to give meaning and then also UMAP which provides a means of visualising the structure of this data globally. However, there is a very distinct gap that exists in this literature. Even though CLIP has so far been well-validated for multimodal data, there has not been enough evidence to support its use case and relevance for humanitarian response to disasters.

This is why I think there is potential for new work and innovations around the field of humanitarian response to disasters and anything disaster related. My research project will try and fill this gap. I would also like to say that I am not attempting to develop or a new architecture for a Neural Network. What I would like to explore instead is whether there is potential in integrating the zero-shot abilities which was found in the CLIP model with the original structure in the UMAP model and also whether the would also work as an “unsupervised exploration” tool. A successful outcome for myself here in my research project would be to prove the efficacy of a new pipeline in crisis response situation. Which is the ability for responders and humanitarian workers to search for information using natural language as opposed to the search options offered in a pretrained model's checkbox list.

## References:

- Andrienko, G., Andrienko, N., Jankowski, P., Keim, D., Kraak, M. -J., MacEachren, A. and Wrobel, S. (2007). Geovisual analytics for spatial decision support: Setting the research agenda. *International Journal of Geographical Information Science*, 21(8), pp.839–857. doi: <https://doi.org/10.1080/13658810701349011>.
- Alam, F., Ofli, F. and Imran, M. (2018) “CrisisMMD: Multimodal Twitter Datasets from Natural Disasters”, *Proceedings of the International AAAI Conference on Web and Social Media*, 12(1). doi: 10.1609/icwsm.v12i1.14983 [Accessed 29 Dec. 2025].
- McInnes, L., Healy, J. and Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. [online] *arXiv.org*. Available at: <https://arxiv.org/abs/1802.0342> [Accessed 29 Dec. 2025].
- Ofli, F., Alam, F. and Imran, M. (2020). Analysis of Social Media Data using Multimodal Deep Learning for Disaster Response. [online] *arXiv.org*. Available at: <https://arxiv.org/abs/2004.11838> [Accessed 30 Dec. 2025].
- Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G. and Sutskever, I. (2021b). Learning Transferable Visual Models From Natural Language Supervision. [online] *proceedings.mlr.press*. Available at: <https://proceedings.mlr.press/v139/radford21a> [Accessed 30 Dec. 2025].
- Castillo, C., 2016. *Big crisis data: social media in disasters and time-critical situations*. Cambridge University Press.
- Imran, M., Castillo, C., Diaz, F. and Vieweg, S., 2015. Processing social media messages in mass emergency: A survey. *ACM computing surveys (CSUR)*, 47(4), pp.1-38.
- Liu, S., Palen, L., Sutton, J., Hughes, A. and Vieweg, S. (2008). On-Line Photo Sharing in Times of Disaster In Search of the Bigger Picture: The Emergent Role of On-Line Photo Sharing in Times of Disaster. [online] Available at: [https://ics.uci.edu/~ucrec/pubs/upload/919\\_Liu2010.pdf](https://ics.uci.edu/~ucrec/pubs/upload/919_Liu2010.pdf) [Accessed 23 Dec. 2025].
- Nguyen, D.T., Ofli, F., Imran, M. and Mitra, P., 2017, July. Damage assessment from social media imagery data during disasters. In *Proceedings of the 2017 IEEE/ACM international conference on advances in social networks analysis and mining 2017* (pp. 569-576).
- Voulodimos, A., Doulamis, N., Doulamis, A. and Protopapadakis, E. (2018). Deep Learning for Computer Vision: a Brief Review. *Computational Intelligence and Neuroscience*, 2018(1), pp.1–13. doi: <https://doi.org/10.1155/2018/7068349>